
CAPSTONE PROJECT

AI STUDY BUDDY

Presented By:

Konchada. Shyam sai – SRM IST, KTR – CINTEL department

OUTLINE

- **Problem Statement** (Should not include solution)
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment (Step by Step Procedure)**
- **Result**
- **Conclusion**
- **Future Scope(Optional)**
- **References**

PROBLEM STATEMENT

-  Problem Statement
- Students lack personalized, 24/7 academic support for instant doubt resolution
- Traditional tutoring is expensive, time-bound, and not always accessible
- Need for adaptive learning assistance across multiple subjects and difficulty levels
- Limited availability of interactive study tools for modern learning needs

SYSTEM APPROACH

Frontend

Streamlit app for user interaction (chat UI, input/output display)

Backend

Python logic to handle chat flow, model inference, and session state

Model Engine

Gemini flash 2.5 based model from google ai developers(e.g., via HuggingFace Transformers or llama.cpp)

Data Handling

Optional logging, feedback collection, or prompt templates

Architecture: Client-Server with real-time streaming

Key Features: Chat interface, multi-subject support, adjustable difficulty levels

Deployment: Render cloud platform

ALGORITHM & DEPLOYMENT

1. Privacy & Security

User conversations must be confidential and not stored without consent.

No personally identifiable information (PII) should be collected or exposed.

Secure communication using HTTPS.

2. Availability

The chatbot should be accessible 24/7 to support users during emotional distress.

Minimal downtime during updates or maintenance.

3. Performance

Response time should be under 2 seconds for typical queries.

Efficient memory usage to support low-resource devices (especially if using LLaMA locally).

4. Usability

Clean, intuitive interface with empathetic tone and easy navigation.

Mobile-responsive design for accessibility on phones and tablets.

Use of icons and emotional cues to guide user interaction.

5. Scalability

Should support increasing user load without degradation in performance.

Modular architecture to allow future integration (e.g., journaling, voice input, multilingual support).

RESULT

Study Tools

Choose your study subject:

General

Explanation Level:

Intermediate

Beginner

Advanced

Study Session Context

Current study topic (optional):

e.g., Calculus, World History, Python

Clear Chat History

Session Stats

Messages exchanged: 5

About

The matrix A is called the **matrix representation** of T with respect to the standard bases.

4. Matrices:

- A **matrix** is a rectangular array of numbers arranged in rows and columns. An $m \times n$ matrix has m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

- Matrix Addition:** Adding two matrices A and B (of the same dimensions) results in a new matrix whose entries are the sum of the corresponding entries:

$$C = A + B \quad \text{where} \quad c_{ij} = a_{ij} + b_{ij}$$

- Scalar Multiplication (Matrix):** Multiplying a matrix A by a scalar c scales each entry of the matrix:

$$B = c \cdot A \quad \text{where} \quad b_{ij} = c \cdot a_{ij}$$

Ask your study question...

Study Tools

Choose your study subject:

General

Explanation Level:

Intermediate

Beginner

Advanced

Study Session Context

Current study topic (optional):

e.g., Calculus, World History, Python

Clear Chat History

Session Stats

Messages exchanged: 3

About

Example prompts:

- "Explain quantum physics like I'm 10 years old"
- "Create a study plan for learning Python in 2 weeks"
- "Give me 5 practice questions about World War II"
- "Help me understand calculus derivatives"
- "Break down the water cycle step by step"

Hi I'm your AI Study Buddy. I'm here to help you learn and understand any subject. What would you like to study today?

Hi

Hi there! How can I help you today?

Ask your study question...

Study Tools

Choose your study subject:

History

Explanation Level:

Beginner

Beginner

Advanced

Study Session Context

Current study topic (optional):

e.g., Calculus, World History, Python

Clear Chat History

Session Stats

Messages exchanged: 7

About

tell me about ancient greece

Ancient Greece was a civilization that flourished in southeastern Europe, primarily on the Greek peninsula and the Aegean islands, from roughly 800 BCE to 600 CE. It left an indelible mark on Western civilization, influencing everything from politics and philosophy to art and architecture. Here's a breakdown of key aspects:

Timeline and Periods:

- Minoan & Mycenaean Civilizations (c. 2700 - 1100 BCE):** These were precursors to what we typically think of as Ancient Greece. The Minoans, based on Crete, were a sophisticated maritime culture. The Mycenaeans, centered on the Peloponnese peninsula, were more warlike and likely responsible for the Trojan War (though the historicity of that event is debated).
- Dark Ages (c. 1100 - 800 BCE):** A period of decline and upheaval following the collapse of the Mycenaean civilization. Written records are scarce.
- Archaic Period (c. 800 - 500 BCE):** Re-emergence of civilization, development of the Greek alphabet, the rise of the polis (city-state), colonization, and the development of early philosophy and literature.
- Classical Period (c. 500 - 323 BCE):** The golden age of Greece. Significant developments in

Ask your study question...

Study Tools

Choose your study subject:

General

Explanation Level:

Intermediate

Beginner

Advanced

Study Session Context

Current study topic (optional):

e.g., Calculus, World History, Python

Clear Chat History

Session Stats

Messages exchanged: 5

About

tell me about linear algebra with math equations

Okay, let's dive into linear algebra with mathematical equations. Linear algebra is fundamentally about vector spaces and linear transformations between them.

1. Vectors and Scalars:

- A **vector** is an object that has magnitude and direction. In linear algebra, we often represent vectors as ordered lists of numbers. For example, in 2D space:

$$v = [v_1, v_2]$$

Where v_1 and v_2 are the components of the vector v in 2D space:

$$v = [v_1, v_2, v_3]$$

More generally, in n -dimensional space:

$$v = [v_1, v_2, \dots, v_n]$$

Ask your study question...

Study Tools

Choose your study subject:

General

Mathematics

Science

History

Programming

Languages

Business

Arts

Other

Session Stats

Messages exchanged: 5

About

Properties of Determinants:

- $\det(A^T) = \det(A)$
- $\det(AB) = \det(A) \cdot \det(B)$
- A matrix is invertible if and only if $\det(A) \neq 0$.

In Summary:

Linear algebra provides a powerful set of tools for working with vectors, matrices, and linear transformations. It is fundamental to many areas of mathematics, science, engineering, and computer science, including:

- Solving systems of equations.
- Data analysis and machine learning.
- Computer graphics and image processing.
- Physics (e.g., mechanics, electromagnetism, quantum mechanics).
- Optimization.

This overview provides a foundation. Each of these topics can be explored in much greater depth. I recommend finding a good textbook or online course to delve deeper into the fascinating world of linear algebra.

Ask your study question...

GITHUB AND DEPLOYMNET LINK

- Attach your Github Link : coding files GITHUB
<https://github.com/shysai/AI-study-buddy.git>
- Deployment link: GITHUB/SOME OTHER PLATFORM = Render.com
- <https://ai-study-buddy-2-usc5.onrender.com/>

CONCLUSION

- The development of an AI Study Buddy using Gemini and Streamlit represents a significant advancement in accessible, personalized education. By combining cutting-edge AI with an intuitive interface, the system provides students with instant academic support, detailed explanations, and adaptive learning assistance—making quality education available to everyone, anytime.

FUTURE SCOPE(OPTIONAL)

- Add Voice Support** Users can talk to the chatbot and hear replies—great for accessibility.
- Support More Languages** Include Tamil and other local languages so more students can use it.
- Work Without Internet** Make it run offline using lightweight models like LLaMA for rural areas.
- Track Mood and Feelings** Let users log their emotions and see patterns over time.
- Connect to Real Help** If someone is in crisis, guide them to a counselor or helpline.

REFERENCES

- List and cite relevant sources, research papers, and articles that were instrumental in developing the proposed solution.
- Streamlit Documentation
Streamlit Inc.
<https://docs.streamlit.io/>
Used for building the interactive web interface and real-time chat components
- Google Generative AI Python SDK
Google AI
https://ai.google.dev/tutorials/python_quickstart
Primary API for integrating Gemini Flash 2.5 AI capabilities
- Render Deployment Documentation
Render Technologies
<https://render.com/docs>
Guidance for cloud deployment and service configuration
- Python-dotenv Package
Saurabh Kumar
<https://github.com/theskumar/python-dotenv>
Secure environment variable management for local development



THANK YOU